Customer Churn Analysis

End-to-End Data Analytics Project using Python, SQL & Power BI

Executive Summary

This project analyzes telecom customer churn using Python, SQL, and Power BI. The objective was to identify churn drivers, high-risk customers, revenue loss, and retention opportunities. The workflow covered data cleaning, exploratory data analysis, business analysis, dashboard development, and insight generation.

Dataset Overview

The dataset contains 7,043 customer records and 35 columns. It includes customer demographics, subscription details, tenure, contract type, internet service, payment method, monthly charges, total charges, churn score, and churn status.

Data Cleaning

Missing value analysis revealed that churn_reason contained null values for active customers. Duplicate analysis confirmed zero duplicate records. Data types were validated and transformed where required. Feature engineering was performed to create tenure groups and customer segments.

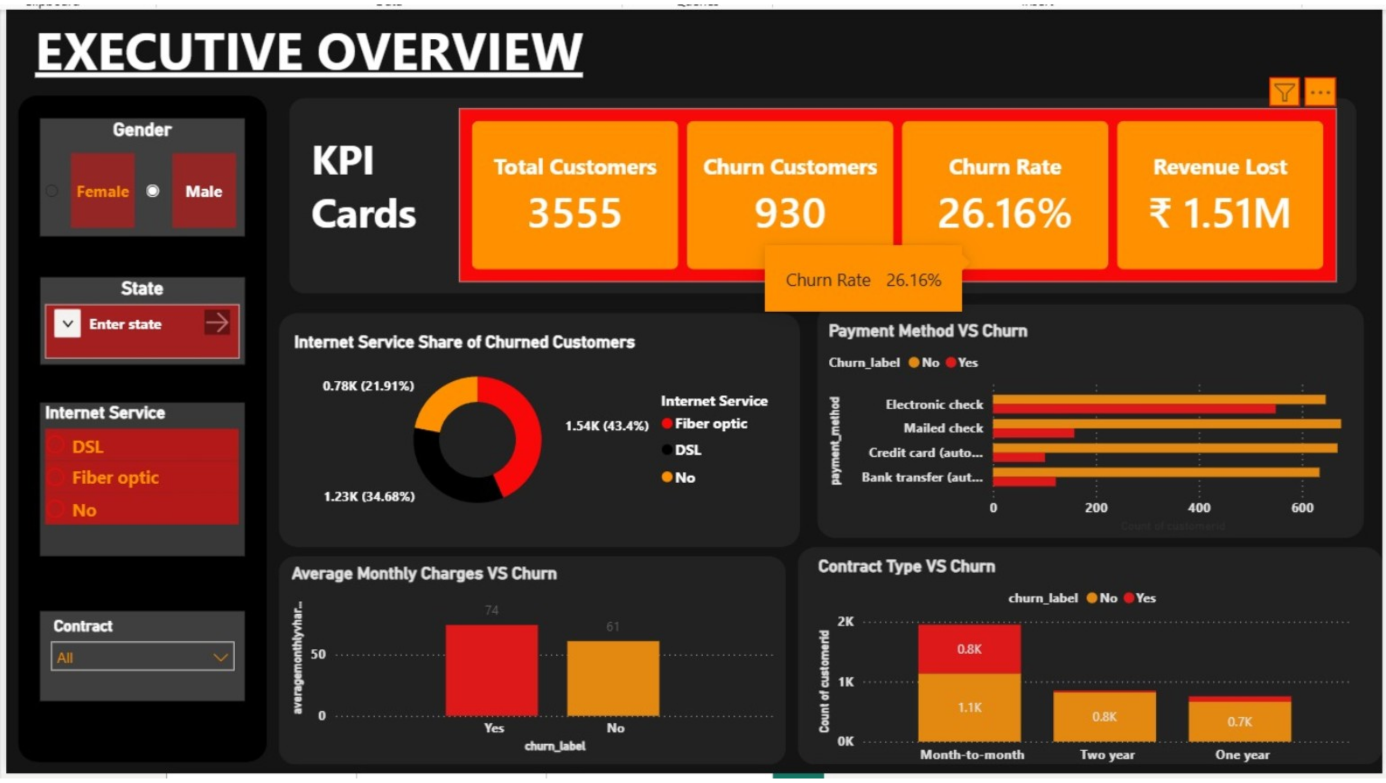
EDA Findings

Customers with shorter tenure were more likely to churn. Customers paying higher monthly charges showed higher churn rates. Month-to-month contracts had significantly more churn than yearly contracts. Electronic check users exhibited the highest churn behavior.

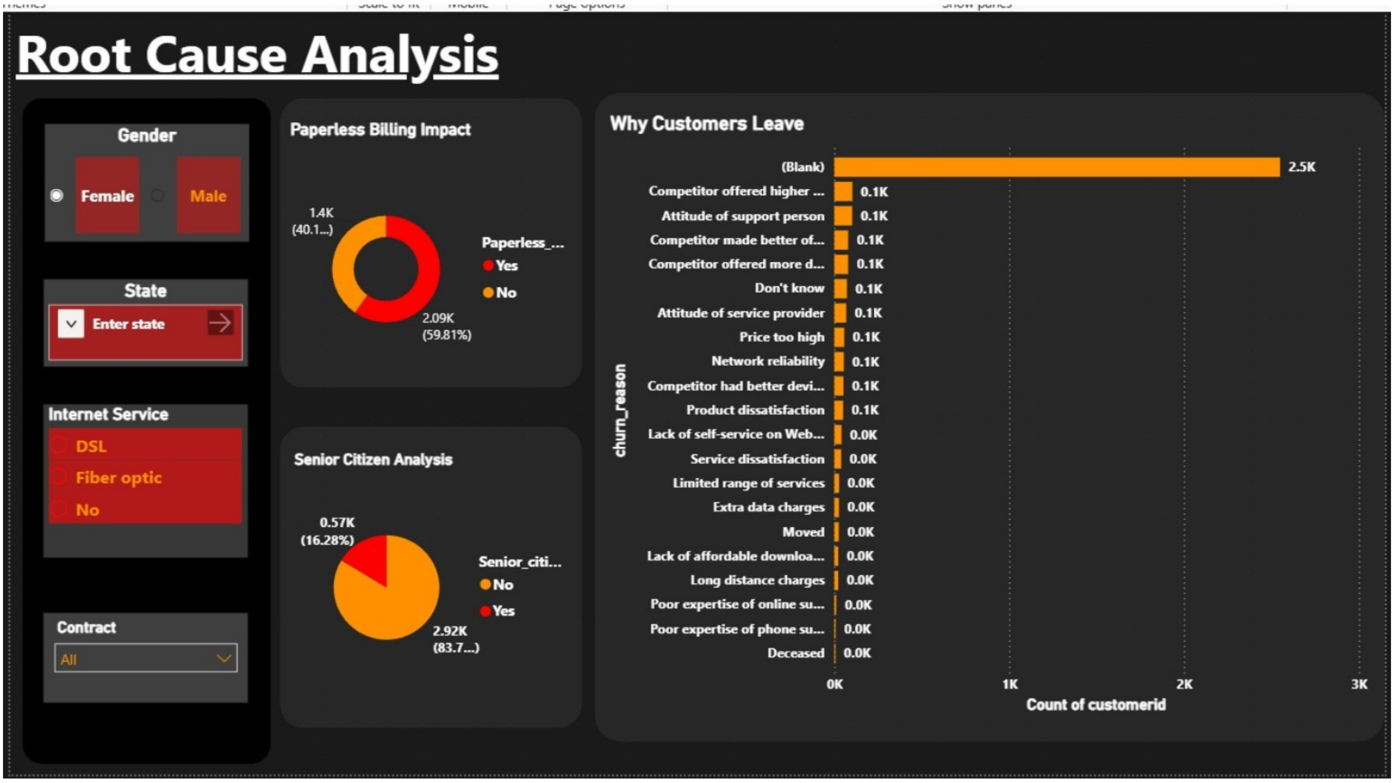
SQL Analysis

SQL queries were used to calculate churn rate, revenue impact, customer segmentation, payment-method analysis, contract-wise churn, churn reason analysis, and high-risk customer identification.

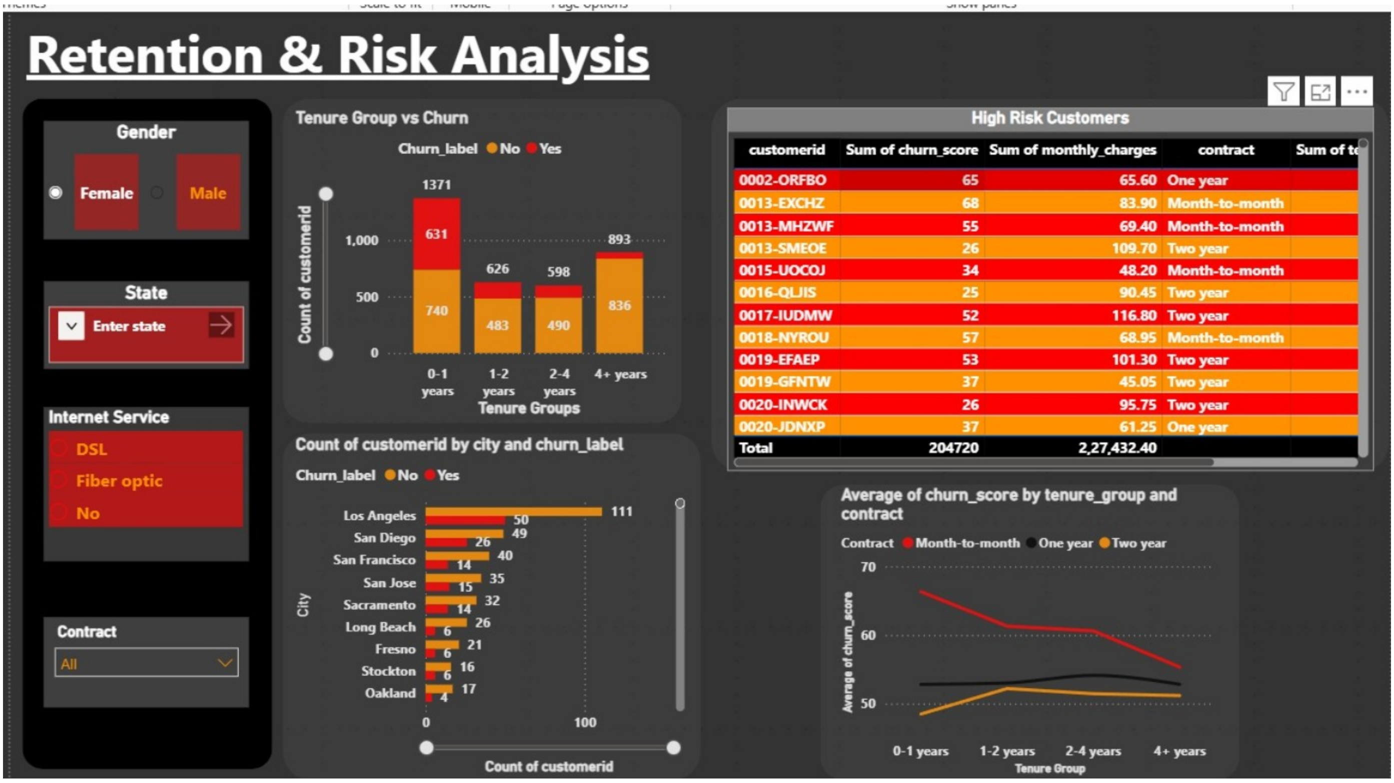
Executive Overview Dashboard



Root Cause Analysis Dashboard



Retention & Risk Analysis Dashboard



Key Findings

- Overall churn rate ≈ 26.5%.
- Month-to-month customers are the most vulnerable segment.
- Higher monthly charges are associated with greater churn risk.
- Fiber optic customers contribute heavily to churn volume.
- Electronic check users show elevated churn behavior.
- Short-tenure customers are more likely to leave.

Business Recommendations

Promote long-term contracts, target high-risk customers with retention campaigns, improve onboarding for new customers, optimize pricing strategies, strengthen customer support, and use predictive analytics for proactive churn prevention.

Conclusion

This project demonstrates an end-to-end data analytics workflow and converts raw telecom customer data into actionable business insights that support customer retention and revenue growth.